



Faculty of Computer Science and Mathematics
Universiti Malaysia Terengganu

LAB REPORT

CSE3023-Web-Based Application Development

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Lab Report	7
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TASKS 1-4:

CODE:

register.html

```
<!DOCTYPE html>
<html>
<head><title>Student Registration</title></head>
<body>
  <h2>Register New Student</h2>
  <form action="UserServlet" method="POST" enctype="multipart/form-data">
    <input type="hidden" name="action" value="register">
    Matric No: <input type="text" name="matricNo" required><br><br>
    Full Name: <input type="text" name="fullname" required><br><br>
    Password: <input type="password" name="password" required><br><br>
    Profile Picture: <input type="file" name="profileImage" accept="image/*"
required><br><br>
    <input type="submit" value="Register">
  </form>
</body>
</html>
```

login.html

```
<!DOCTYPE html>
<html>
<body>
  <h2>Student Login</h2>
  <form action="UserServlet" method="POST">
    <input type="hidden" name="action" value="login">
    Matric No: <input type="text" name="matricNo" required><br><br>
    Password: <input type="password" name="password" required><br><br>
    <input type="submit" value="Login">
  </form>
  <br>
  <a href="register.html">Don't have an account? Register here</a>
</body>
</html>
```

dashboard.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@page import="com.lab.bean.StudentBean"%>
```

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Student Dashboard</title>
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
  <style>
    .profile-img {
      width: 160px;
      height: 160px;
      object-fit: cover;
      border: 5px solid #198754;
    }
  </style>
</head>
<body class="bg-light">
  <%
    response.setContentType("text/html;charset=UTF-8");
    StudentBean user = (StudentBean) session.getAttribute("loggedUser");
    if (user == null) {
      response.sendRedirect("login.html");
      return;
    }
  %>
  <div class="container d-flex justify-content-center align-items-center" style="min-
height: 100vh;">
    <div class="card shadow-lg border-0 rounded-4 p-4 p-md-5 text-center"
style="max-width: 600px; width: 100%;">
      <div class="card-body">
        
        <h2 class="fw-bold text-dark mb-2">Welcome, <%=
user.getFullname() %>!</h2>
        <p class="text-muted fs-5 mb-4">Matric Number: <strong class="text-
secondary"><%= user.getMatricNo() %></strong></p>

        <div class="d-flex justify-content-center gap-3 flex-wrap">
          <a href="SubjectServlet?action=view" class="btn btn-primary px-4 py-2
fw-semibold">Manage Subjects</a>

          <a href="UserServlet?action=logout" class="btn btn-dark px-4 py-2 fw-
semibold">Logout</a>
          <a href="UserServlet?action=delete" onclick="return
confirm('WARNING: Are you sure you want to delete your account permanently?');"
class="btn btn-danger px-4 py-2 fw-semibold">Delete Account</a>
        </div>
      </div>
    </div>
  </div>

```

```
        </div>
    </div>
    <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></
script>
</body>
</html>
```

StudentBean.java

```
package com.lab.bean;

public class StudentBean implements java.io.Serializable {
    private String matricNo;
    private String fullname;
    private String password;
    private String base64Image;

    public StudentBean() {}

    public String getMatricNo() { return matricNo; }
    public void setMatricNo(String matricNo) { this.matricNo = matricNo; }

    public String getFullname() { return fullname; }
    public void setFullname(String fullname) { this.fullname = fullname; }

    public String getPassword() { return password; }
    public void setPassword(String password) { this.password = password; }

    public String getBase64Image() { return base64Image; }
    public void setBase64Image(String base64Image) { this.base64Image =
base64Image; }
}
```

UserServlet.java

```
package com.lab.controller;

import com.lab.bean.StudentBean;
import com.lab.dao.StudentDAO;
import java.io.IOException;
import java.io.InputStream;
import javax.servlet.ServletException;
import javax.servlet.annotation.MultipartConfig;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
```

```

import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;
import javax.servlet.http.Part;

@MultipartConfig(maxFileSize = 16177215) // max 16 MB required for image
upload
public class UserServlet extends HttpServlet {
    private StudentDAO studentDAO = new StudentDAO();

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        String action = request.getParameter("action");

        if ("register".equals(action)) {
            StudentBean newStudent = new StudentBean();
            newStudent.setMatricNo(request.getParameter("matricNo"));
            newStudent.setFullname(request.getParameter("fullname"));
            newStudent.setPassword(request.getParameter("password"));

            InputStream inputStream = null;
            Part filePart = request.getPart("profileImage");
            if (filePart != null && filePart.getSize() > 0) {
                inputStream = filePart.getInputStream();
            }

            boolean isSuccess = studentDAO.registerStudent(newStudent, inputStream);
            if (isSuccess) {
                response.getWriter().println("Registration Successful! <a
href='login.html'>Login here</a>");
            } else {
                response.getWriter().println("Registration Failed!");
            }
        } else if ("login".equals(action)) {
            String matric = request.getParameter("matricNo");
            String pass = request.getParameter("password");
            StudentBean student = studentDAO.loginStudent(matric, pass);

            if (student != null) {
                HttpSession session = request.getSession();
                session.setAttribute("loggedUser", student);
                response.sendRedirect("dashboard.jsp");
            } else {
                response.getWriter().println("Invalid Credentials! <a href='login.html'>Try
Again</a>");
            }
        }
    }
}

```

```

@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
    String action = request.getParameter("action");
    HttpSession session = request.getSession(false);

    if ("logout".equals(action)) {
        if (session != null) {
            session.invalidate();
        }
        response.sendRedirect("login.html");
    } else if ("delete".equals(action)) {
        if (session != null && session.getAttribute("loggedUser") != null) {
            StudentBean user = (StudentBean) session.getAttribute("loggedUser");
            studentDAO.deleteStudent(user.getMatricNo());
            session.invalidate();
        }
        response.sendRedirect("register.html");
    }
}
}

```

StudentDAO.java

```

package com.lab.dao;

import com.lab.bean.StudentBean;
import java.io.ByteArrayOutputStream;
import java.io.InputStream;
import java.sql.Blob;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Base64;

public class StudentDAO {
    private Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return DriverManager.getConnection("jdbc:mysql://localhost:3306/lab7_db",
"root", "admin");
    }

    public boolean registerStudent(StudentBean student, InputStream imageStream) {
        try (Connection conn = getConnection()) {

```

```

        String sql = "INSERT INTO students (matric_no, password, fullname,
profile_image) VALUES (?, ?, ?, ?)";
        PreparedStatement pstmt = conn.prepareStatement(sql);
        pstmt.setString(1, student.getMatricNo());
        pstmt.setString(2, student.getPassword());
        pstmt.setString(3, student.getFullname());
        pstmt.setBlob(4, imageStream);
        return pstmt.executeUpdate() > 0;
    } catch (Exception e) {
        e.printStackTrace();
        return false;
    }
}

public StudentBean loginStudent(String matricNo, String password) {
    StudentBean student = null;
    try (Connection conn = getConnection()) {
        String sql = "SELECT * FROM students WHERE matric_no = ? AND
password = ?";
        PreparedStatement pstmt = conn.prepareStatement(sql);
        pstmt.setString(1, matricNo);
        pstmt.setString(2, password);
        ResultSet rs = pstmt.executeQuery();

        if (rs.next()) {
            student = new StudentBean();
            student.setMatricNo(rs.getString("matric_no"));
            student.setFullname(rs.getString("fullname"));

            Blob blob = rs.getBlob("profile_image");
            if (blob != null) {
                InputStream inputStream = blob.getBinaryStream();
                ByteArrayOutputStream outputStream = new ByteArrayOutputStream();
                byte[] buffer = new byte[4096];
                int bytesRead = -1;
                while ((bytesRead = inputStream.read(buffer)) != -1) {
                    outputStream.write(buffer, 0, bytesRead);
                }
                byte[] imageBytes = outputStream.toByteArray();

                student.setBase64Image(Base64.getEncoder().encodeToString(imageBytes));
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
        return student;
    }

    public boolean deleteStudent(String matricNo) {

```

```

try (Connection conn = getConnection()) {
    String sql = "DELETE FROM students WHERE matric_no = ?";
    PreparedStatement pstmt = conn.prepareStatement(sql);
    pstmt.setString(1, matricNo);
    return pstmt.executeUpdate() > 0;
} catch (Exception e) {
    e.printStackTrace();
    return false;
}
}
}

```

OUTPUT:

Register New Student

Matric No:

Full Name:

Password:

Profile Picture: IKHWAN.jfif

← ↻ ⓘ localhost:8080/LabExercise7/UserServlet

Registration Successful! [Login here](#)

Student Login

Matric No:

Password:

Login

[Don't have an account? Register here](#)



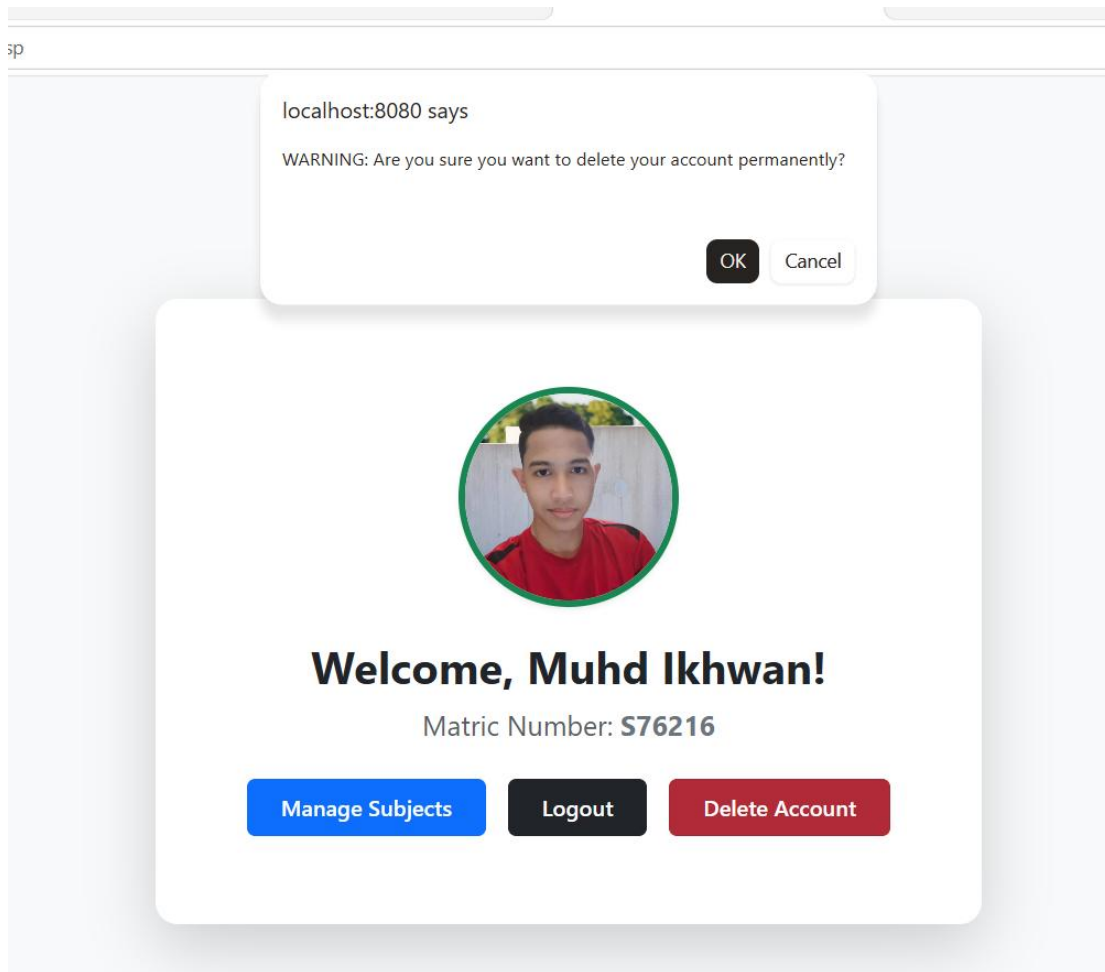
Welcome, Muhd Ikhwan!

Matric Number: **S76216**

Manage Subjects

Logout

Delete Account



Reflections

1. Why is it highly recommended to clear the Output log before testing a specific feature that might contain bugs?

Clearing the output log provides a clean slate ensuring that any errors you see are directly caused by your most recent actions. This makes it much easier to pinpoint the exact line of code causing a new bug, without wasting time getting confused by old, irrelevant error messages.

2. Why is it considered a bad practice in a production environment to display the full `e.printStackTrace()` error message directly on the user's web browser?

Displaying a full stack trace is a major security risk because it exposes sensitive backend details like database names and file paths that hackers can exploit. Additionally, it creates a poor user experience by confusing regular visitors with unreadable code instead of providing a friendly, helpful error page.

EXERCISE 7.8:

CODE:

viewSubject.jsp

```
<%@page import="java.util.List"%>
<%@page import="com.lab.bean.SubjectBean"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head><title>My Registered Subjects</title></head>
<body>
    <h2>Subject List</h2>
    <a href="<%= request.getContextPath() %>/SubjectServlet?action=new">Register
New Subject</a> |
    <a href="<%= request.getContextPath() %>/dashboard.jsp">Back to
Dashboard</a>
    <br><br>
    <table border="1">
        <tr>
            <th>ID</th>
            <th>Subject Code</th>
            <th>Subject Name</th>
            <th>Actions</th>
        </tr>
        <%
            List<SubjectBean> subjects = (List<SubjectBean>)
request.getAttribute("listSubjects");
            if (subjects != null && !subjects.isEmpty()) {
                for (SubjectBean sub : subjects) {
                    %>
                    <tr>
                        <td><%= sub.getId() %></td>
                        <td><%= sub.getSubject_code() %></td>
                        <td><%= sub.getSubject_name() %></td>
                        <td>
                            <a href="<%=
request.getContextPath() %>/SubjectServlet?action=edit&id=<%=
sub.getId() %>">Edit</a> |
                            <a href="<%=
request.getContextPath() %>/SubjectServlet?action=delete&id=<%= sub.getId() %>"
                                onclick="return confirm('Are you sure you want to delete this
subject?');">Delete</a>
                        </td>
                    </tr>
                <%
            }
        %>
```

```

        }
    } else {
    %>
    <tr>
        <td colspan="4">No subjects registered yet.</td>
    </tr>
    <% } %>
</table>
</body>
</html>

```

updateSubject.jsp

```

<%@page import="com.lab.bean.SubjectBean"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head><title>Update Subject</title></head>
<body>
    <h2>Edit Subject</h2>
    <% SubjectBean subject = (SubjectBean) request.getAttribute("subject"); %>
    <form action="<%= request.getContextPath() %>/SubjectServlet" method="post">
        <input type="hidden" name="action" value="update">
        <input type="hidden" name="id" value="<%= subject.getId() %>">
        <label>Subject Code:</label>
        <input type="text" name="subject_code" value="<%=
subject.getSubject_code() %>" required><br><br>
        <label>Subject Name:</label>
        <input type="text" name="subject_name" value="<%=
subject.getSubject_name() %>" required><br><br>
        <input type="submit" value="Update">
        <a href="<%=
request.getContextPath() %>/SubjectServlet?action=view">Cancel</a>
    </form>
</body>
</html>

```

registerSubject.jsp

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head><title>Register Subject</title></head>
<body>
    <h2>Register New Subject</h2>
    <form action="<%= request.getContextPath() %>/SubjectServlet" method="post">
        <input type="hidden" name="action" value="add">
        <label>Subject Code:</label>
        <input type="text" name="subject_code" required><br><br>
        <label>Subject Name:</label>

```

```

        <input type="text" name="subject_name" required><br><br>
        <input type="submit" value="Register">
        <a href="%=
request.getContextPath() %>/SubjectServlet?action=view">Cancel</a>
    </form>
</body>
</html>

```

SubjectServlet.java

```

package com.lab.controller;

import com.lab.bean.SubjectBean;
import com.lab.bean.StudentBean;
import com.lab.dao.SubjectDAO;
import java.io.IOException;
import java.util.List;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

public class SubjectServlet extends HttpServlet {
    private SubjectDAO subjectDAO;

    public void init() {
        subjectDAO = new SubjectDAO();
    }

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        processRequest(request, response);
    }

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        processRequest(request, response);
    }

    private void processRequest(HttpServletRequest request, HttpServletResponse
        response) throws ServletException, IOException {
        HttpSession session = request.getSession(false);
        if (session == null || session.getAttribute("loggedUser") == null) {
            response.sendRedirect("login.html");
            return;
        }

        StudentBean loggedUser = (StudentBean) session.getAttribute("loggedUser");
        String matricNo = loggedUser.getMatricNo();
    }

```

```

String action = request.getParameter("action");
if (action == null) { action = "view"; }

switch (action) {
    case "new": showNewForm(request, response); break;
    case "add": insertSubject(request, response, matricNo); break;
    case "edit": showEditForm(request, response); break;
    case "update": updateSubject(request, response, matricNo); break;
    case "delete": deleteSubject(request, response, matricNo); break;
    case "view":
    default: listSubjects(request, response, matricNo); break;
}
}

private void showNewForm(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
    request.getRequestDispatcher("/subject/registerSubject.jsp").forward(request,
response);
}

private void showEditForm(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
    int id = Integer.parseInt(request.getParameter("id"));
    SubjectBean existingSubject = subjectDAO.selectSubjectById(id);
    request.setAttribute("subject", existingSubject);
    request.getRequestDispatcher("/subject/updateSubject.jsp").forward(request,
response);
}

private void insertSubject(HttpServletRequest request, HttpServletResponse
response, String matricNo) throws IOException {
    String code = request.getParameter("subject_code");
    String name = request.getParameter("subject_name");
    SubjectBean newSubject = new SubjectBean(matricNo, code, name);
    subjectDAO.insertSubject(newSubject);
    response.sendRedirect("SubjectServlet?action=view");
}

private void updateSubject(HttpServletRequest request, HttpServletResponse
response, String matricNo) throws IOException {
    int id = Integer.parseInt(request.getParameter("id"));
    String code = request.getParameter("subject_code");
    String name = request.getParameter("subject_name");
    SubjectBean subject = new SubjectBean(id, matricNo, code, name);
    subjectDAO.updateSubject(subject);
    response.sendRedirect("SubjectServlet?action=view");
}

```

```

private void deleteSubject(HttpServletRequest request, HttpServletResponse
response, String matricNo) throws IOException {
    int id = Integer.parseInt(request.getParameter("id"));
    subjectDAO.deleteSubject(id, matricNo);
    response.sendRedirect("SubjectServlet?action=view");
}

private void listSubjects(HttpServletRequest request, HttpServletResponse
response, String matricNo) throws ServletException, IOException {
    List<SubjectBean> listSubjects =
subjectDAO.selectAllSubjectsByMatric(matricNo);
    request.setAttribute("listSubjects", listSubjects);
    request.getRequestDispatcher("/subject/viewSubjects.jsp").forward(request,
response);
}
}

```

SubjectDAO.java

```

package com.lab.dao;

import com.lab.bean.SubjectBean;
import java.sql.*;
import java.util.ArrayList;
import java.util.List;

public class SubjectDAO {
    protected Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return DriverManager.getConnection("jdbc:mysql://localhost:3306/lab7_db",
"root", "admin");
    }

    public void insertSubject(SubjectBean subject) {
        String sql = "INSERT INTO registered_subjects (matric_no, subject_code,
subject_name) VALUES (?, ?, ?)";
        try (Connection conn = getConnection(); PreparedStatement stmt =
conn.prepareStatement(sql)) {
            stmt.setString(1, subject.getMatric_no());
            stmt.setString(2, subject.getSubject_code());
            stmt.setString(3, subject.getSubject_name());
            stmt.executeUpdate();
        } catch (Exception e) { e.printStackTrace(); }
    }

    public List<SubjectBean> selectAllSubjectsByMatric(String matric_no) {
        List<SubjectBean> subjects = new ArrayList<>();
        String sql = "SELECT * FROM registered_subjects WHERE matric_no = ?";
        try (Connection conn = getConnection(); PreparedStatement stmt =
conn.prepareStatement(sql)) {

```

```

        stmt.setString(1, matric_no);
        ResultSet rs = stmt.executeQuery();
        while (rs.next()) {
            int id = rs.getInt("id");
            String subject_code = rs.getString("subject_code");
            String subject_name = rs.getString("subject_name");
            subjects.add(new SubjectBean(id, matric_no, subject_code, subject_name));
        }
    } catch (Exception e) { e.printStackTrace(); }
    return subjects;
}

public SubjectBean selectSubjectById(int id) {
    SubjectBean subject = null;
    String sql = "SELECT * FROM registered_subjects WHERE id = ?";
    try (Connection conn = getConnection(); PreparedStatement stmt =
conn.prepareStatement(sql)) {
        stmt.setInt(1, id);
        ResultSet rs = stmt.executeQuery();
        if (rs.next()) {
            String matric_no = rs.getString("matric_no");
            String subject_code = rs.getString("subject_code");
            String subject_name = rs.getString("subject_name");
            subject = new SubjectBean(id, matric_no, subject_code, subject_name);
        }
    } catch (Exception e) { e.printStackTrace(); }
    return subject;
}

public void updateSubject(SubjectBean subject) {
    String sql = "UPDATE registered_subjects SET subject_code = ?, subject_name
= ? WHERE id = ? AND matric_no = ?";
    try (Connection conn = getConnection(); PreparedStatement stmt =
conn.prepareStatement(sql)) {
        stmt.setString(1, subject.getSubject_code());
        stmt.setString(2, subject.getSubject_name());
        stmt.setInt(3, subject.getId());
        stmt.setString(4, subject.getMatric_no());
        stmt.executeUpdate();
    } catch (Exception e) { e.printStackTrace(); }
}

public void deleteSubject(int id, String matric_no) {
    String sql = "DELETE FROM registered_subjects WHERE id = ? AND
matric_no = ?";
    try (Connection conn = getConnection(); PreparedStatement stmt =
conn.prepareStatement(sql)) {
        stmt.setInt(1, id);
        stmt.setString(2, matric_no);
        stmt.executeUpdate();
    }
}

```



```

        } catch (Exception e) { e.printStackTrace(); }
    }
}

```

SubjectBean.java

```
package com.lab.bean;
```

```

public class SubjectBean implements java.io.Serializable {
    private int id;
    private String matric_no;
    private String subject_code;
    private String subject_name;

    public SubjectBean() {}

    public SubjectBean(String matric_no, String subject_code, String subject_name) {
        this.matric_no = matric_no;
        this.subject_code = subject_code;
        this.subject_name = subject_name;
    }

    public SubjectBean(int id, String matric_no, String subject_code, String
subject_name) {
        this.id = id;
        this.matric_no = matric_no;
        this.subject_code = subject_code;
        this.subject_name = subject_name;
    }

    public int getId() { return id; }
    public void setId(int id) { this.id = id; }

    public String getMatric_no() { return matric_no; }
    public void setMatric_no(String matric_no) { this.matric_no = matric_no; }

    public String getSubject_code() { return subject_code; }
    public void setSubject_code(String subject_code) { this.subject_code =
subject_code; }

    public String getSubject_name() { return subject_name; }
    public void setSubject_name(String subject_name) { this.subject_name =
subject_name; }
}

```

web.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<web-app version="4.0" xmlns="http://xmlns.jcp.org/xml/ns/javaee"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

```

```
xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee
http://xmlns.jcp.org/xml/ns/javaee/web-app_4_0.xsd">
  <servlet>
    <servlet-name>SubjectServlet</servlet-name>
    <servlet-class>com.lab.controller.SubjectServlet</servlet-class>
  </servlet>
  <servlet>
    <servlet-name>StudentDAO</servlet-name>
    <servlet-class>com.lab.dao.StudentDAO</servlet-class>
  </servlet>
  <servlet>
    <servlet-name>UserServlet</servlet-name>
    <servlet-class>com.lab.controller.UserServlet</servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name>SubjectServlet</servlet-name>
    <url-pattern>/SubjectServlet</url-pattern>
  </servlet-mapping>
  <servlet-mapping>
    <servlet-name>StudentDAO</servlet-name>
    <url-pattern>/StudentDAO</url-pattern>
  </servlet-mapping>
  <servlet-mapping>
    <servlet-name>UserServlet</servlet-name>
    <url-pattern>/UserServlet</url-pattern>
  </servlet-mapping>
  <session-config>
    <session-timeout>
      30
    </session-timeout>
  </session-config>
</web-app>
```

OUTPUT:

Register New Subject

Subject Code:

Subject Name:

[Cancel](#)

Subject List

[Register New Subject](#) | [Back to Dashboard](#)

ID	Subject Code	Subject Name	Actions
1	CSF3023	WEB2	Edit Delete

Edit Subject

Subject Code:

Subject Name:

[Cancel](#)

>

Subject List

[Register New Subject](#) | [Back to Dashboard](#)

ID	Subject Code	Subject Name	Actions
1	CSF3023	WEB DEVELOPMENT	Edit Delete

localhost:8080/LabExercise7/SubjectServlet?action=view

Subject List

[Register New Subject](#) | [Back to Dashboard](#)

ID	Subject Code	Subject Name	Actions
1	CSF3023	WEB DEVELOPMENT	Edit Delete

localhost:8080 says

Are you sure you want to delete this subject?

OK

Cancel

Subject List

[Register New Subject](#) | [Back to Dashboard](#)

ID	Subject Code	Subject Name	Actions
No subjects registered yet.			